

Umer Farooq

HPC & AI INFRASTRUCTURE ENGINEER, AND SOLUTION ARCHITECT

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SUMMARY

I build the systems underneath AI and HPC workloads — the bare-metal Kubernetes and HPC platforms they run on, the GPU kernels that make them fast, and the pipelines that check their own output before returning it. The work runs in three strands: infrastructure and platform engineering for enterprise GPU-as-a-Service deployments, the solution architecture and support models behind the bids for them, and the GPU, quantum and AI development the platforms exist to serve. Every number on this site is one I measured.

EXPERIENCE

HPC & infrastructure engineer

Jun 2026 – present

MAK Technology · Dubai, United Arab Emirates

- Built the control plane that replaced manual SSH and ad-hoc playbooks on a production HPC cluster: 15 orchestrated tasks over four host tiers, with live log streaming and OIDC-gated RBAC.
- Ran a nine-part security and performance audit of a delivered platform and remediated every critical finding before handover — 5 critical and 1 high closed out of 20 tracked.
- Design and automate bare-metal Kubernetes and HPC platforms for enterprise GPU-as-a-Service deployments — zero-touch provisioning, hardened cluster bootstrap, GitOps delivery, and Slurm/GPFS scheduling over parallel storage.

Kubernetes · Ansible · Kubespray · MAAS · ArgoCD · Terraform · Slurm · GPFS · Keycloak · Prometheus · Python · FastAPI

HPC & quantum software engineer

Sep 2025 – Aug 2026

Open Quantum Workbench, FAST-NUCES · Islamabad, Pakistan

- Built the GPU-accelerated and distributed side: hybrid MPI+OpenMP tensor-network simulation with CUDA contractions, for circuits past 20 qubits.
- Built the browser-side simulation interface and the Python compute services behind it, so a circuit could be run and inspected without a local quantum toolchain.
- Put Qiskit, Cirq, and PennyLane behind one interface, with OpenQASM 3.0 as the exchange format between them, and scheduled the simulation jobs on Kubernetes.

Python · FastAPI · React · Qiskit · PennyLane · OpenQASM 3.0 · Kubernetes · CUDA · MPI · OpenMP

Freelance developer — Level 2 seller

Aug 2023 – Aug 2024

Fiverr · Remote

- Delivered 30+ full-stack applications on the MERN stack and .NET, reaching roughly 500 end users.
- Wrote 2D games in C++ with SFML and SDL2 — custom physics, collision detection, and a rendering path 30% faster than the first working version.
- Built desktop line-of-business applications in C# and Java over PostgreSQL and MySQL, MVC throughout, sized for 100+ concurrent users.
- Closed 100+ projects at 98% client satisfaction and 80% repeat custom, delivering about 20% ahead of the agreed date.

React · Node.js · Express · MongoDB · PostgreSQL · C++ · C# · Java · .NET · SFML · SDL2

Core team member — PR and marketing

Feb – May 2025

NaSCon'25, FAST-NUCES · Islamabad, Pakistan

- Managed community partners at 15+ universities: inter-university meetings, RSVP tracking, and campaign content across LinkedIn, Instagram, and Facebook.
- Ran partner-facing promotion for one of Pakistan's largest student-run technology conferences.

Officer — communications and outreach

Feb – May 2025

NaSCon'25, FAST-NUCES · Islamabad, Pakistan

- Handled correspondence between FAST-NUCES Islamabad and partner universities, and confirmed speaker and sponsor invitations.

EDUCATION

Bachelor of Computer Science

Aug 2022 – Jun 2026

National University of Computer and Emerging Sciences (FAST-NUCES)

- Dean's List Award – Spring 2023

COURSEWORK

High-Performance Computing · Parallel & Distributed Computing · Compiler Construction · Operating Systems · Advanced Programming · Data Structures & Algorithms · Database Systems · Computer Networks · Artificial Intelligence · Machine Learning

SELECTED PROJECTS · 8

HPC Cluster Management Platform

2026

A FastAPI and Celery control plane where every long-running operation has one shape — route, task, a single SSH session, Redis ledger, WebSocket — so a reload reattaches instead of losing the run.

CRITICAL FINDINGS CLOSED 5 from 20 tracked ORCHESTRATION SURFACE 15 tasks
SCHEDULER MEMORY ACCURACY ~93% MANUAL SSH OPERATIONS 0 from every change

Python · FastAPI · Celery · Redis · React · asyncssh

Bare-Metal Kubernetes Automation Platform

2026

Idempotent Kubespray/Ansible playbooks over a version-controlled inventory, with hardening applied inside the bootstrap run so the cluster is never briefly unhardened.

MANUAL OS INSTALL STEPS 0 from every node CLUSTER & ADD-ON STATE IN GIT 100%
CONTROL-PLANE REPLICAS 3× DAY-2 RUNBOOKS AS AUTOMATION 5

Ansible · Kubespray · Helm · YAML · Kubernetes

GPU Cloud Reference Architecture

2026

How the isolation is actually enforced: tenant-scoped control-plane APIs, admission policy that blocks on violation, per-tenant KMS keys, and quota objects with defaults rather than conventions.

TENANT ISOLATION LAYERS 5 ARCHITECTURE DIAGRAMS AUTHORED ~50 TENANT-DEDICATED HARDWARE 0

Kubernetes · OPA/Gatekeeper · OIDC · Terraform

Enterprise SLA & Incident-Response Framework

2026

Where the framework touches the system: the monitoring that detects, the runbooks that respond, and the preventive changes shipped through the same GitOps path as everything else.

P0 RESPONSE COMMITMENT 15 min P0 RESTORE TARGET 4 h SEVERITY TIERS AND ESCALATION LEVELS 4 × 4
SERVICE REVIEW CADENCE Monthly

Prometheus · Loki · ArgoCD · Terraform

Cirq-RAG Code Assistant

Generative AI & RAG

Quantum code generation without hallucinations

CIRCUIT GENERATION SUCCESS 92% from 52% KNOWLEDGE BASE ENTRIES 140+

Python · Cirq · RAG · FAISS · Multi-Agent · Prompt Engineering

QCanvas

Mar 2025 – Present

Unified Quantum Simulator Platform

3rd Prize · Huawei ICT Competition National Finals (UniQ Team)

Next.js · TypeScript · FastAPI · Qiskit · Cirq · PennyLane · OpenQASM

Parallel Tensor Network Quantum Simulator (Q-Tensor)

Academic/Research

Hybrid MPI/OpenMP + CUDA Quantum Simulation

C++ · CUDA · OpenMP · MPI · METIS · CMake · Quantum Computing

MNIST Classification with GPU Acceleration

High-Performance Computing & GPU

Deep Learning Optimization

CUDA · Python · PyTorch · Nsight · Tensor Cores · FP16 Mixed Precision

26 further projects — compilers, distributed hash tables, desktop applications, and games — are listed in full on the work page.

ALL 34 PROJECTS →

SKILLS

HIGH-PERFORMANCE COMPUTING & PARALLELISM

GPU computing with CUDA, Tensor Cores, mixed-precision training

OpenMP, MPI, and hybrid MPI+OpenMP parallel programming

Performance profiling and optimization with NVIDIA Nsight, perf, and Valgrind

MACHINE LEARNING, DEEP LEARNING & LLM SYSTEMS

PyTorch, TensorFlow/Keras for CNNs, RNNs/LSTMs, diffusion and GAN models

Retrieval-Augmented Generation (RAG), semantic search, FAISS, evaluation with NDCG/MAP

Multi-agent AI workflows, prompt engineering, and tool-augmented pipelines

QUANTUM COMPUTING & SIMULATION

Qiskit, Cirq, PennyLane, and OpenQASM 3.0

Quantum circuit simulation, tensor networks, and hybrid classical-quantum workflows

SYSTEMS, DISTRIBUTED & NETWORKING

Modern C/C++ for systems programming and algorithmic optimization

Distributed hash tables, IPFS-style storage, P2P and client-server architectures

Socket programming (TCP/UDP), multithreading with pthreads, concurrency control

FULL-STACK DEVELOPMENT & PLATFORMS

React, Next.js, Node.js/Express, FastAPI, RESTful API design

.NET, JavaFX, desktop applications with PostgreSQL/MySQL backends

Kubernetes, Docker, GitHub Actions, and CI/CD for reproducible deployments

INFRASTRUCTURE, ORCHESTRATION & AUTOMATION

Bare-metal provisioning with MAAS, Warewulf diskless imaging, and IPMI/Redfish out-of-band control

Hardened Kubernetes bootstrap with Kubespray and Ansible — encrypted CNI, Pod Security Standards, least-privilege RBAC, CIS-aligned kubelet

GitOps delivery with ArgoCD, Helm and Kustomize; Terraform and inventory-as-code

Slurm scheduling and GPFS parallel storage; Spack and Lmod environment stacks

Prometheus, Grafana and Loki observability; Keycloak OIDC single sign-on and role-based access

SOLUTION ARCHITECTURE & TECHNICAL PRE-SALES

Enterprise reference architectures and RFP-grade technical proposals for large-scale bids

Multi-tenant isolation design across identity, quota, network, GPU/MIG partitioning and data

SLA and incident-response frameworks — severity tiers, response and restore commitments, escalation ladders, and the RCA loop back into runbooks

Compliance, RACI and architecture diagram sets that hold up under customer review

CERTIFICATIONS

2025 Oracle Cloud Infrastructure Certified Generative AI Professional Oracle CREDENTIAL

AWARDS

- 2025 3rd prize — Huawei ICT Competition, national finals With the UniQ team, for QCanvas.
- 2023 Dean's List Spring 2023, FAST-NUCES.
- 2024 Level 2 seller, Fiverr 100+ projects delivered at 98% satisfaction.
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LANGUAGES

Urdu NATIVE OR BILINGUAL English NATIVE OR BILINGUAL Japanese ELEMENTARY
